

The relationship between Li ion conductivity and crystal structure for ordered perovskite compounds, $(\text{La}_{2/3-1/3p}\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ ($p=0.05, 0.11$ and 0.14)

Tetsuhiro Katsumata*, Megumi Takahata, Nobuko Mochizuki, Yoshiyuki Inaguma

Department of Chemistry, Faculty of Science, Gakushuin University, 1-5-1 Mejiro, Toshima, Tokyo 171-8588, Japan

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Abstract

We synthesized furnace-cooled $(\text{La}_{2/3-1/3p}\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ (LLMW) and quenched $(\text{La}_{2/3-1/3p}\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ (LLMWq) and investigated Li ion conductivity for these compounds. Activation energy was found to decrease with Li content for LLMW and LLMWq, while the obvious lattice contraction accompanied with Li content was not observed. For LLMW, ordered arrangement of the A-site ions was varied with Li content which seems to be responsible for the decrease in the activation energy. As results of structure refinements of LLMWq, it was revealed that covalent character of W–O bond increased with Li content. It is considered that the Coulomb repulsion between W and Li ions dominates the activation energy and that the increase in the covalent character of the W–O bond with Li content is the reason for the decrease in the activation energy.

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1. Introduction

Much attention has been paid to perovskite-type Li ion conductors for both application and fundamental researches. In particular, $\text{La}_{0.55}\text{Li}_{0.35}\text{TiO}_3$ is one of the solid Li ion conductors showing the highest Li ion conductivity, 10^{-3} S/cm, at room temperature [1,2]. In this compound, Li ions, La ions and vacancies are distributed in the A-site, and the Li ion, which is smaller than the La ion, can easily migrate through the conduction path formed by the vacancies and Li ions in the structure [3].

For perovskite-type oxides, ABO_3 , many B-site complex perovskites, which have two different B ions with different valences, have been synthesized. In the case of perovskite-type Li ion conductors, various B-site complex perovskites have also been synthesized and investigated their Li ion conductivity, especially the relation between the activation

energy and the lattice parameter and/or the valence of the B ion [4–11]. Inaguma et al. [4] have investigated the dependence of the activation energy on the lattice parameters for perovskite-type Li ion conductors having a single B ion as well as two different B ions. Their results include data that can be divided into two groups, one composed of compounds containing only the Ti ion as B-site ion and the other composed of compounds containing the octavalent cation (Nb^{5+} and Ta^{5+}) as B-site ion. While the activation energy has been found to increase linearly with decreases in the lattice parameters for both groups, the activation energy in the latter group is higher than that in the former group when the lattice parameters are the same. Furthermore, Harada et al. [5] have prepared many perovskite-type Li ion conductors and have investigated the relationship between ionic conductivity and the lattice parameters. These compounds can be classified into five series of solid solutions:

- (A) $(\text{La}_{1-p}\text{Nd}_p)_{0.56}\text{Li}_{0.33}\text{TiO}_3$,
- (B) $\text{La}_{0.56}\text{Li}_{0.33}\text{M}_p^{4+}\text{Ti}_{1-p}\text{O}_3$ ($\text{M}^{4+} = \text{Zr, Hf}$),
- (C) $(\text{Ca}_{1-p}\text{Sr}_p)_{0.56}\text{Li}_{0.33}\text{Ta}_{0.56}\text{Ti}_{0.44}\text{O}_3$,

* Corresponding author. Tel.: +81-3-3986-0221; fax: +81-3-5992-1029.

E-mail address: 20001996@gakushuin.ac.jp (T. Katsumata).

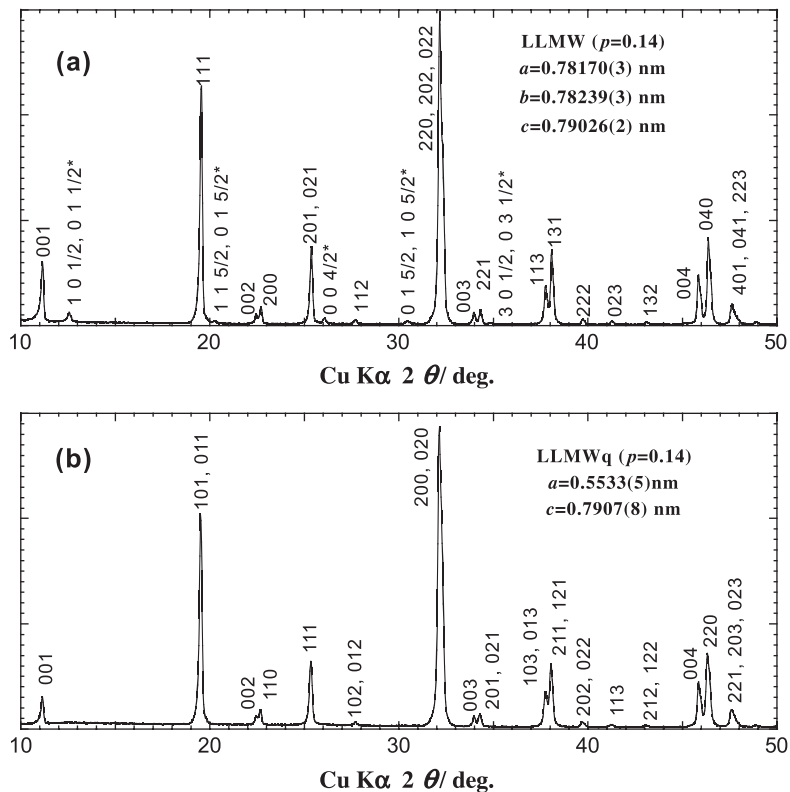


Fig. 1. Powder X-ray diffraction patterns of $p=0.14$ for LLMW (a) and LLMWq (b). The miller indices (*) in panel 1 correspond to the super lattice lines indexed using $2a' \times 2b' \times 4c'$ multiple cell.

- (D) $(\text{Ca}_{1-p}\text{Sr}_p)_{0.56}\text{Li}_{0.33}\text{Fe}_{0.225}\text{Ta}_{0.775}\text{O}_3$, and
 (E) $\text{Sr}_{0.56}\text{Li}_{0.33}\text{M}^{3+}_{0.225}\text{Ta}_{0.775}\text{O}_3$ ($\text{M}^{3+} = \text{Cr}, \text{Fe}, \text{Co}, \text{Ga}, \text{Y}$).

The variations in ionic conductivity can also be divided into two groups, one composed of the series (A) and (B), containing only tetravalent ions in the B-site, and the other composed of the series (C), (D), and (E), containing the Ta ion in the B-site. The ionic conductivity was found to reach a maximum at 0.387 and 0.395 nm, respectively, for the two groups. According to earlier studies, Li ion conduction is influenced primarily by higher-valence B-site ions for B-site complex perovskites. However, dominant influence of the higher-valence cation has not yet been explained.

On the other hand, in our preliminary work [6], the synthesis of a perovskite-type lithium ion conductor $(\text{La}_{0.63}\text{Li}_{0.1})(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ containing Mg^{2+} and W^{6+} as

B-site ions was successful, and it was found that the resulting compound has ordered arrangements of B-sites (Mg and W ions are in the NaCl-type ordered arrangement) and A-sites (La ions are in alternatively ordered arrangement along one axis), the same as $\text{La}_{2/3}(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ [12]. In the present study, in order to investigate the influence of the valence of B-site ions in complex B-site perovskites, we synthesized perovskite-type Li conductors, $(\text{La}_{2/3-p}\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ (LLMW; $p=0.05, 0.11$ and 0.14) and investigated the variation of the Li ion conductivity with p .

2. Experimental

These compounds were synthesized by a conventional solid-state reaction technique. La_2O_3 (4 N), Li_2CO_3 (3 N), WO_3 (4 N) and MgO (3 N) were used as starting materials. The metal content of La_2O_3 and MgO were analyzed by

Table 1
The lattice parameters and perovskite parameters for LLMW

p	a (nm)	b (nm)	c (nm)	a_p (nm)
0.00[12]	0.78157(5)	0.78344(6)	0.79067(6)	0.392161(3)
0.05	0.7824(9)	0.7835(9)	0.7906(6)	0.3927(3)
0.11	0.7811(6)	0.7833(4)	0.7912(3)	0.3926(4)
0.14	0.78170(3)	0.78329(3)	0.79029(2)	0.39253(2)

Table 2
The lattice parameters and perovskite parameters for LLMWq

x	a (nm)	c (nm)	a_p (nm)
0.05	0.5533(5)	0.7903(6)	0.3925(2)
0.11	0.5534(6)	0.7903(9)	0.3926(2)
0.14	0.5533(5)	0.7903(8)	0.3926(2)

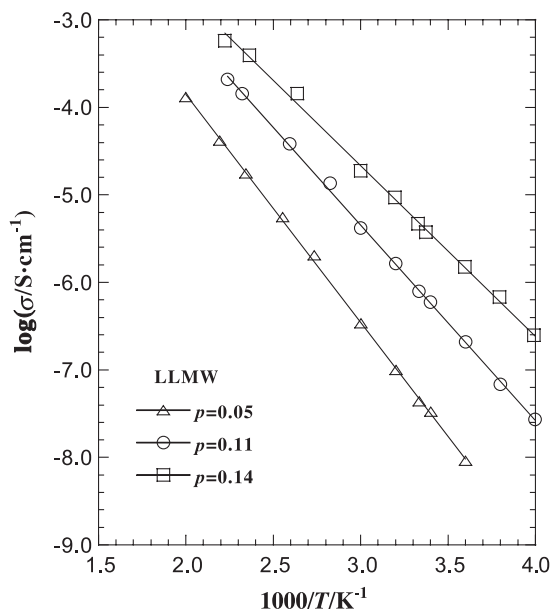


Fig. 2. Arrhenius plot of LLMW [$p=0.05$ (Δ), 0.11 ($\circ$) and 0.14 ($\square$)].

chelation with ethylenediaminetetraacetic acid (EDTA). The stoichiometric mixture of these materials was ground and pelletized, and the pellet was then calcined at 1373 K for 12 h; resulting pellet was then reground, pelletized and calcined again at 1373 K for 12 h. After this pellet was reground and pressed into pellets with a diameter of 7 mm and a thickness of approximately 1.5 mm, it was sintered at 1423 K for 6 h and was cooled in the furnace. These samples are referred to as LLMW. Furthermore, this sintered sample was heated at 1373 K for 30 min and quenched in the liquid nitrogen. These samples are referred to as $(\text{La}_{2/3} - 1/3p\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ (LLMWq).

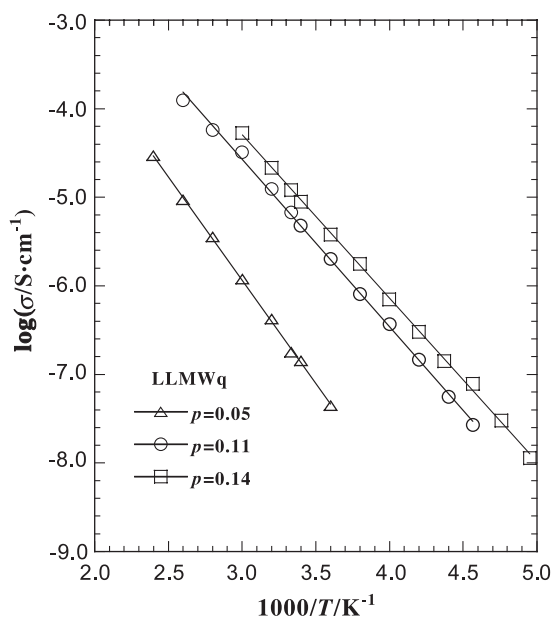


Fig. 3. Arrhenius plot of LLMWq [$p=0.05$ (Δ), 0.11 ($\circ$) and 0.14 ($\square$)].

Table 3

The Li ion conductivity σ and the activation energy E for LLMW and LLMWq

p	LLMW		LLMWq	
	σ_{300} (S/cm)	E (eV)	σ_{300} (S/cm)	E (eV)
0.05	4.4×10^{-8}	0.52	1.8×10^{-7}	0.46
0.11	7.9×10^{-7}	0.44	6.8×10^{-6}	0.38
0.14	4.7×10^{-6}	0.39	1.2×10^{-5}	0.37

The phase identification and the determination of the lattice parameters was carried out by a powder X-ray diffraction method (XRD; Rigaku RINT 2100). The structure refinement was executed by the Rietveld method. The powder-diffraction data were corrected at room temperature in the 2θ range of 10° – 120° , with a step size of 0.02° and for 2 s per point. The refinement was performed using program RIETAN 2000 [13].

The Au electrode was pasted at 727 K and was evaporated for the LLMW and LLMWq, respectively. The ionic conductivity was measured by an ac impedance method from 250 to 550 K. For the measurement from 250 to 400 K, a precision LCR meter (Agilent technology 4284A) was used in the frequency range of 20 Hz–1 MHz. Measuring from 400 to 550 K, we used the impedance analyzer (Yokogawa Hewlett Packard 4192A) in the frequency range of 5 Hz–13 MHz.

3. Results and discussion

A single phase was obtained for $p=0.05$, 0.11 and 0.14 of LLMW and LLMWq. Fig. 1a shows the X-ray diffrac-

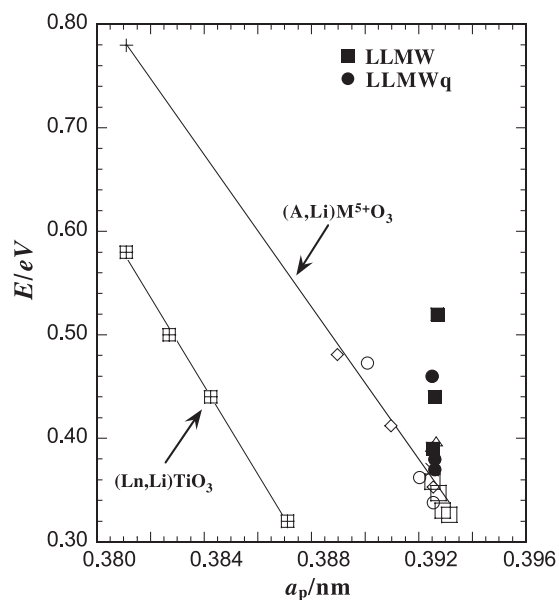


Fig. 4. Relationship between the activation energy and the perovskite parameter for $\text{Ln}_{0.5}\text{Li}_{0.5}\text{TiO}_3$ ($\text{Ln}=\text{La}, \text{Pr}, \text{Nd}$ and Sm ; $\boxplus$), $0.5\text{CaTiO}_3 - 0.5\text{LiTaO}_3$ ($+$), $\text{La}_x\text{Ag}_y\text{Li}_{1-x-y}\text{NbO}_3$ ($\circ$), $\text{La}_x\text{Na}_y\text{Li}_{1-x-y}\text{NbO}_3$ ($\diamond$), $\text{SrTiO}_3 - \text{LiTaO}_3$ ($\square$), $x\text{SrTiO}_3 - (1-x)\text{LiNbO}_3$ (Δ) and $\text{La}_{0.25}\text{Na}_{0.20}\text{Li}_{0.05}\text{TaO}_3$ ($\times$).

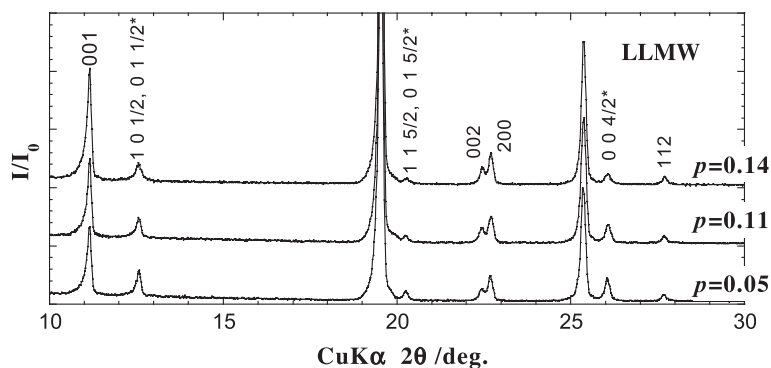


Fig. 5. Comparison of super structure lines for LLMW. The miller indices and the miller indices (*) correspond to the superstructure line for doubling and quadrupling ordering, respectively.

tion patterns of $p=0.14$ for LLMW. For LLMW, the diffraction peaks can be indexed by the $2a' \times 2b' \times 2c'$ orthorhombic cell, where a' , b' and c' are cell parameters of the perovskite subcell. The structure of LLMW is found to be an analogue for the mother compound, $\text{La}_{2/3}(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ [12]. The superlattice lines indexed using the $2a' \times 2b' \times 2c'$ multiple cell are caused by the NaCl-type ordering of W^{6+} and Mg^{2+} ions at the B-site and by the alternative arrangement of La and Li ions and vacancies along the c -axis. While the reflection peaks indexed by the $2a' \times 2b' \times 4c'$ multiple cell were also observed for $\text{La}_{2/3}(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$, the ordered arrangement which requires a quadrupling has not been clarified. Fig. 1b shows the X-ray diffraction patterns of $p=0.14$ for LLMWq. The superlattice lines corresponding to a $2a' \times 2b' \times 4c'$ multiple cell were diminished for LLMWq. In the case of LLMWq, the unit cell is $\sqrt{2}a' \times \sqrt{2}b' \times 2c'$ tetragonal, as the apparent peak separation between ($h00$) and ($00l$) was not observed, and a better refinement by Rietveld analysis was not

obtained supposing that a tilting model of a BO_6 octahedron using an orthorhombic cell.

Tables 1 and 2 show the lattice parameters, the perovskite parameters, a_p , (square root of the volume of perovskite unit cell) for LLMW and LLMWq, respectively. While a_p slightly decreased with increases in the Li content for LLMW, the a_p of LLMWq is constant for all samples within the experimental error.

Figs. 2 and 3 shows the Arrhenius plots of the Li ion conductivity for LLMW and LLMWq, respectively. Good linearity was observed for this temperature range. Table 3 shows the ionic conductivity at 300 K, σ_{300} and the activation energy E of LLMW and LLMWq. The highest ionic conductivity, 1.2×10^{-5} S/cm, was obtained for $p=0.14$ of LLMWq.

As shown in Table 3, the E is decreased with the increase in p , even though the a_p values for both LLMW and LLMWq were the same. These behaviors are quite different from those of other perovskite-type Li ion conductors. Fig. 4 shows the variation of the activation energy with the lattice

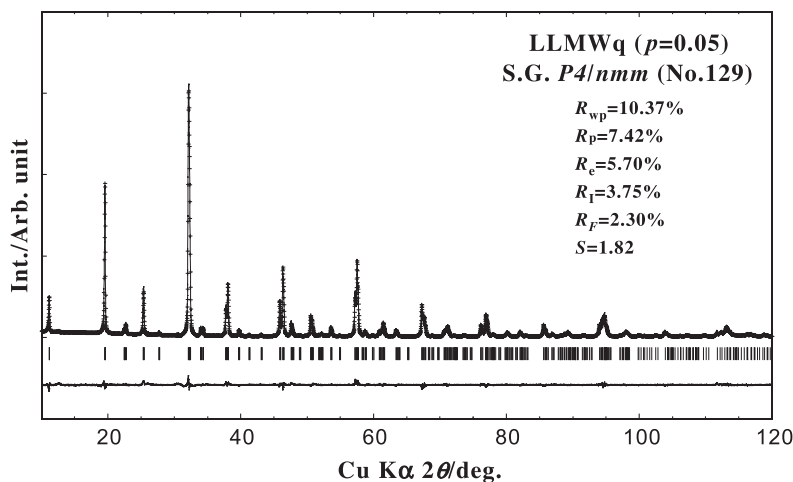


Fig. 6. The observed (upper solid line) and calculated (+) X-ray diffraction patterns, the peak position (|), the difference between observed and calculated patterns (lower solid line) for LLMWq($p=0.05$).

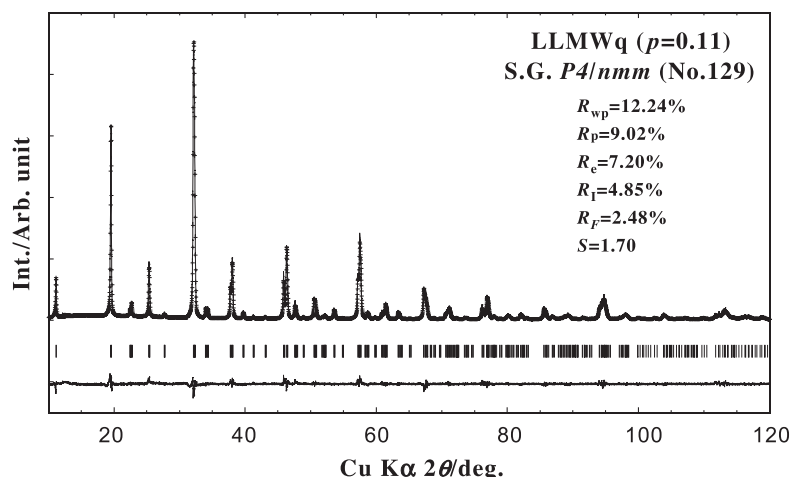


Fig. 7. The observed (upper solid line) and calculated (+) X-ray diffraction patterns, the peak position (|), the difference between observed and calculated patterns (lower solid line) for LLMWq ($p=0.11$).

parameter for various perovskite-type Li ion conductors. As mentioned above, the activation energy of perovskite-type Li ion conductors often linearly increased with the decrease in the lattice, and this dependence is influenced by the valence of the B-site ion; that is, the activation energy of the compound having only Ti ion at B-site is smaller than that of the compound having the octavalent ion at the B-site when the lattice parameters are the same. Therefore, the activation energy of LLMW and LLMWq is expected to not vary with Li content and to be much larger than that of the compound having octavalent cation because both compounds contains the W^{6+} ion as the B-site ion. These experimental results indicate that the activation energy of LLMW and LLMWq is influenced by other factors except for the lattice parameter.

For LLMW, it is suggested that the change in the ordered arrangement influences the variations in the activation energy. Fig. 5 provides a comparison of the relative

intensities of the X-ray diffraction patterns for LLMW. The intensities of the superstructure diffraction peaks indexed by the $2a' \times 2b' \times 2c'$ multiple cell and the $2a' \times 2b' \times 4c'$ multiple cell increased and decreased with Li content, respectively. These findings indicate that the arrangement of ordering, primarily ordering of the La ion varies with Li content; accordingly, the decrease in the activation energy is responsible for the variation in the ordered arrangement.

For LLMWq, no obvious change in the super lattice lines was observed, thus some other factor would influences the activation energy. To investigate the crystal structure of LLMWq in more detail, the Rietveld analysis was executed. As shown in Fig. 1a, the reflection condition was $h+k=2n$ for $(hk0)$ planes. Furthermore, all reflections for $(00l)$ and $(0kl)$ planes were observed. Hence, the possible space group is $P4/n$ (No. 85) or $P4/nmm$ (No. 129). In this study, we adopted the space group having higher symmetry, $P4/nmm$

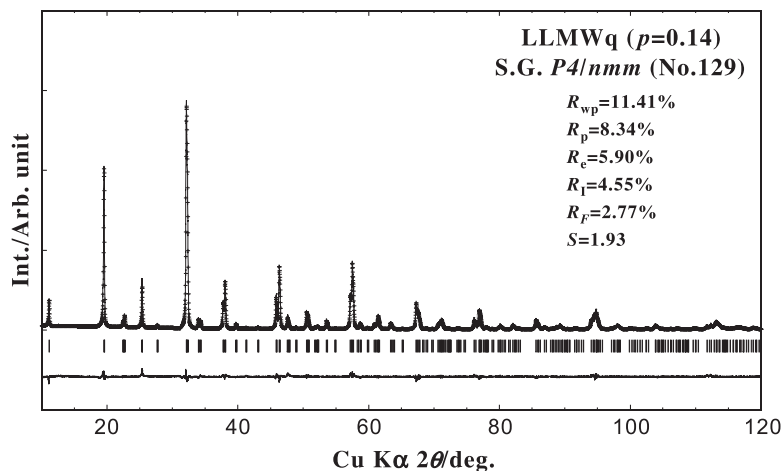


Fig. 8. The observed (upper solid line) and calculated (+) X-ray diffraction patterns, the peak position (|), the difference between observed and calculated patterns (lower solid line) for LLMWq ($p=0.14$).

Table 4
Structural parameters for LLMWq ($p=0.05$)

Site		g	x	y	z	$B/\text{\AA}^2$
La1	2a	0.965(5)	0	0	0	0.1(1)
La2	2b	0.335 ^a	0	0	1/2	1.1(3)
Li	2b	0.1	0	0	1/2	0.5
W	2c	1.0	0	1/2	0.2642(5)	0.05(7)
Mg	2c	1.0	0	1/2	0.738(3)	0.7(4)
O1	8j	1.0	0.247(2)	0.747 ^b	0.245(2)	1.0
O2	2c	1.0	0	1/2	−0.007(5)	1.0
O3	2c	1.0	0	1/2	0.491(5)	1.0

S.G. $P4/nmm$ (No. 129); $a=0.5533(5)$ nm; $c=0.7903(6)$ nm; $R_{wp}=10.37\%$; $R_p=7.42\%$; $R_e=5.70\%$; $S=1.82$; $R_1=3.75\%$; $R_F=2.30\%$.

^a $g(\text{La}_2)=1.3-g(\text{La}_1)$.

^b $y(\text{O}_1)=x(\text{O}_1)+1/2$.

(No. 129). Figs. 6, 7 and 8 show the X-ray diffraction patterns and the Rietveld refinement of LLMWq for $p=0.05$, 0.11 and 0.14, respectively. For all samples, better refinements were obtained. The reason for the somewhat larger R_{wp} being obtained for these refinements is that the quadrupling arrangement of A-site ions was not diminished completely. While the sample was taken from the furnace and put into liquid N_2 as soon as possible, the cooling rate was not enough to completely diminish the arrangement. Tables 4, 5 and 6 shows the crystal data of LLMWq for $p=0.05$, 0.11 and 0.14, respectively. Both W and Mg ions displaced toward the La-poor-layer and the displacement increased with Li content. The occupation of La ions in the La-poor-layer was found to influence the displacement of B-site ions. The O ions in-plane position (O_1 sites) which are called O ions in-plane displaced along the c -axis. However, the direction of displacement for the O ions in-plane changed with the Li content; that is, the O ions in-plane displaced toward La-rich-layer and La-poor-layer for $p=0.05$, 0.11 and $p=0.14$, respectively. The displacement of O ions in plane was found to be influenced by not only the ordered arrangement of A-site ions but also the displacement of B-site ions. Table 7 shows the length of W–O and Mg–O bonds for LLMWq. All oxide ions displaced toward the W ion and this displacement also increased with

Table 5
Structural parameters for LLMWq ($p=0.11$)

Site		g	x	y	z	$B/\text{\AA}^2$
La1	2a	0.946(6)	0	0	0	0.2(1)
La2	2b	0.314 ^a	0	0	1/2	1.4(5)
Li	2b	0.22	0	0	1/2	0.5
W	2c	1.0	0	1/2	0.2646(6)	0.27(9)
Mg	2c	1.0	0	1/2	0.737(4)	0.7(4)
O1	8j	1.0	0.240(2)	0.740 ^b	0.249(2)	1.0
O2	2c	1.0	0	1/2	0.000(6)	1.0
O3	2c	1.0	0	1/2	0.487(6)	1.0

S.G. $P4/nmm$ (No. 129); $a=0.5534(6)$ nm; $c=0.7903(9)$ nm; $R_{wp}=12.24\%$; $R_p=9.02\%$; $R_e=7.20\%$; $S=1.70$; $R_1=4.85\%$; $R_F=2.48\%$.

^a $g(\text{La}_2)=1.26-g(\text{La}_1)$.

^b $y(\text{O}_1)=x(\text{O}_1)+1/2$.

Table 6
Structural parameters for LLMWq ($p=0.14$)

Site		g	x	y	z	$B/\text{\AA}^2$
La1	2a	0.932(6)	0	0	0	0.1(1)
La2	2b	0.308 ^a	0	0	1/2	1.2(3)
Li	2b	0.28	0	0	1/2	0.5
W	2c	1.0	0	1/2	0.2649(5)	0.23(8)
Mg	2c	1.0	0	1/2	0.734(3)	0.7(4)
O1	8j	1.0	0.237(2)	0.737 ^b	0.254(2)	1.0
O2	2c	1.0	0	1/2	0.000(5)	1.0
O3	2c	1.0	0	1/2	0.487(5)	1.0

S.G. $P4/nmm$ (No. 129); $a=0.5533(5)$ nm; $c=0.7903(6)$ nm; $R_{wp}=11.41\%$; $R_p=8.34\%$; $R_e=5.90\%$; $S=1.93$; $R_1=4.55\%$; $R_F=2.77\%$.

^a $g(\text{La}_2)=1.24-g(\text{La}_1)$.

^b $y(\text{O}_1)=x(\text{O}_1)+1/2$.

Li content. The crystal structure of LLMWq ($p=0.05$) is shown in Fig. 9.

For perovskite-type Li ion conductors, it has been considered that the decrease in the bottleneck size causes the increase in the activation energy accompanied with the lattice contraction [14–16]. On the other hand, the molecular dynamics simulation predicts that the covalent character of the B-site ion also affect the Li ion conductivity [17]. To discuss the influence of structure on the Li ion conduction, the bond valence sum and the bottleneck size in the La-poor-layer are estimated for LLMWq.

The bottleneck of LLMWq is defined as the area surrounded by four O ions between A-sites in a perovskite unit cell, and the bottleneck size means the width of this area. Hence, the size of bottleneck in the La-poor layer, S_b , can be estimated from the following equation,

$$S_b = 2c(0.5 - z)a/\sqrt{2} \quad (1)$$

where z is the z coordination of O_1 and a and c are the lattice parameters. Table 8 shows the bottleneck size in the La-poor-layer. As shown in Table 8, the bottleneck size decreases with p . Assuming that the activation energy depends only on the bottleneck size, it would increase with p .

Based on the concept of bond valence [18], the valence of ion i , V_i (in this study, V_i is called the ‘bond

Table 7
Bond lengths of W–O and Mg–O bonds

	$p=0.05$	$p=0.11$	$p=0.14$
W–O1 $\times 4$	1.94(2)	1.88(2)	1.86(2)
W–O2 $\times 1$	2.15(4)	2.09(5)	2.09(4)
W–O3 $\times 1$	1.80(4)	1.76(5)	1.76(4)
W–O(average)	1.95	1.90	1.88
Mg–O1 $\times 4$	1.98(2)	2.04(5)	2.06(2)
Mg–O2 $\times 1$	2.01(5)	2.08(5)	2.10(5)
Mg–O3 $\times 1$	1.95(5)	1.97(5)	1.95(5)
Mg–O(average)	1.98	2.04	2.05

valence sum' in order to distinguish between the valence numbers calculated from the composition and the sum of bond valences) is defined as the sum of all valences of a bond:

$$V_i = \sum_j v_{ij} \quad (2)$$

where v_{ij} is the valence of a bond between ions i and j . The v_{ij} is commonly expressed by

$$v_{ij} = \exp \left[\frac{(R_{ij} - d_{ij})}{b} \right] \quad (3)$$

where d_{ij} , b , and R_{ij} are, respectively, bond length between ions i and j , 'universal' constant equal to 0.37 Å, and the bond valence parameter between ions i and j which is determined empirically. In this study, the R_{ij} of the Mg–O and W–O bonds are 1.693 and 1.921 Å, respectively [18]. Because the bond valence increases with the decrease in the bond length, the greater the covalent character of the bond of the ion pairs is, the larger the bond valence sum is. Table 8 lists the bond valence sum of W and Mg ions for LLMWq. The V_{Mg} is larger than the normal oxidation number and decreases with the Li content, indicating that the Mg–O bond has a covalent character that decreases with the Li content. The V_{W} are larger and smaller than the normal oxidation number for $p=0.11$ and 0.14 and $p=0.05$, respectively, and the V_{W} increased with Li content. These indicate that the W–O bond varies from an ionic character to a covalent character, and that the covalent character increases with the Li content.

The activation energy is influenced by both the short-range repulsion between Li and oxide ions and the Coulomb repulsion between the Li and B-site ions at the bottleneck. For LLMWq, the activation energy is considered to be dominated by the Coulomb repulsion between Li and W ions due to high valence of W ion. The increase in the

Table 8

The average displacement, bottleneck size and bond valence sum for LLMWq

p	Bottleneck size in La-poor-layer (Å ²)	Bond valence sum	
		W	Mg
0.05	15.75	5.7	2.8
0.11	15.52	6.6	2.4
0.14	15.22	6.9	2.3

covalent character of the W ion decreases the effective charge of the W ion, which leads to the decrease in the Coulomb repulsion between the Li and W ions. Therefore, the increase in the covalent character of the W ion decreases the activation energy. This deduction is supported by the experimental results indicating that the Li ion conduction is influenced by the valence of the B-site ion being higher for the complex perovskites. In complex B-site perovskite, in which B-site ions are distributed randomly, the bottleneck near the B-site ion having a higher valence is predicted to be more unstable energetically. Hence, the bottleneck near the B-site ion with higher valence is a saddle point on the conduction path, and the activation energy of B-site complex perovskites is dominated by a B-site ion having higher valence.

4. Conclusion

We synthesized furnace-cooled $(\text{La}_{2/3-1/3p}\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ (LLMW) and quenched $(\text{La}_{2/3-1/3p}\text{Li}_p)(\text{Mg}_{1/2}\text{W}_{1/2})\text{O}_3$ (LLMWq) and investigated the Li ion conductivity for these compounds. It was found that the activation energy decreases with the Li content for LLMW and LLMWq without the lattice contraction, which means that the activation energy is not dominated by the lattice parameter, unlike other perovskite-type Li ion conductors.

For LLMW, the ordered arrangement of the A-site ions was found to vary the Li content. The decrease in activation energy is therefore responsible for this variation in the ordered arrangement.

Based on the Rietveld refinement of LLMWq, it was revealed that the covalent character of the W–O bond increases with the increase in the Li content. The activation energy of LLMWq, mainly influenced by the Coulomb repulsion between Li and W ions due to large valence of W ion and the decrease in the activation energy is responsible for the decrease in the effective charge of the W ion due to an increase in the covalent character of the W–O bond.

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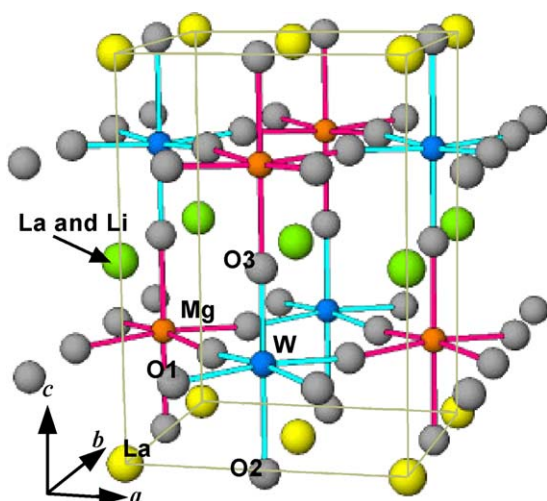


Fig. 9. The crystal structures of LLMWq ($p=0.05$).

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